Complementary Joint Learning for Weakly Supervised Solar Panel Mapping in Aerial Images

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***Abstract*—Classical alternative training schemes for weakly supervised learning (WSL) rely on an initialization-refinement process, which is prone to performance degradation due to the inherent noise and inductive bias in pseudo-labels (PLs). To address this challenge, we propose a novel complementary joint learning framework (CJLF), which leverages the complementary strengths of diverse PL types and substantially reduces the adverse effects of inherent noise and inductive bias in PLs. CJLF employs a shared encoder and a two-stream decoder architecture, with each stream serving a distinct purpose. The object segmentation stream (OSS) is tasked with producing integral predictions with well-defined boundaries, leveraging a cross-scale attention accumulation strategy to resolve object ambiguities in complex scenes. The object localization stream (OLS) aims to identify target objects accurately. To foster the collaboration between these two streams, we propose an attention fusion module (AFM) that integrates self-attention features from OSS and cross-stream attention features. In addition, a Subspace Contrast Constraint is introduced to improve discrimination in ambiguous regions, ensuring OSS robustness against PL noise and strengthening the complementary interaction between the streams. The effectiveness of CJLF is validated on three challenging datasets for solar panel mapping (SPM), including two aerial-image datasets, AerialGDA2020 and DualGMS, and one satellite-image dataset, GloSoFarID. Experimental results, including comparisons with state-of-the-art methods and extensive ablation studies, demonstrate the superiority of CJLF.**

***Index Terms*—Aerial images, complementary learning, object segmentation, solar panel mapping (SPM), weakly supervised learning (WSL).**

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I. Introduction

With the use of solar panels, solar photovoltaic (PV) technology converts readily available solar energy into electricity, reducing dependence on coal-fired power plants and supporting the goal of achieving a carbon-neutral society. Over the past decades, both centralized and decentralized solar PV systems have seen rapid expansion in deployment globally. This rapid growth has heightened public interest in effectively monitoring solar panel installations and assessing their potential social and environmental impacts [1]. With the availability of high-resolution remote sensing images, automatic solar panel mapping (SPM) is feasible with machine learning algorithms [2], [3], [4], which enables consistent and frequent monitoring of solar panel installations for subsequent social-economic analysis.

Following the general workflow of geospatial object segmentation, earlier research primarily relied on traditional classifiers like support vector machines [5] and random forests [4]. Recently, significant advancements have been made with the advent of deep learning techniques. This progress has been largely driven by the use of convolutional neural network (CNN)-based methods combined with densely annotated training data [6], [7], [8], [9]. Although most studies in the field of SPM have utilized end-to-end fully supervised object segmentation models with satisfactory results, it is crucial to recognize that pixel-level labeling under a fully supervised paradigm is both time-consuming and labor-intensive. For remote sensing images, particularly, the annotation process requires more domain-specific knowledge than is typically needed for natural scenes. To address this issue, weakly supervised learning (WSL) has been introduced to alleviate the annotation burdens in acquiring the training data [10], [11].

Detecting solar panels from remote sensing images in a weakly supervised manner, however, is challenging, primarily due to two key factors.

1) *Limited Supervision Provided by Weak Labels:* Due to the lack of annotation details in weak labels, WSL-based methods face challenges in obtaining results with precise and well-defined boundaries, indicating a deficiency in capturing the object variability as well as separating the desired object from the background.

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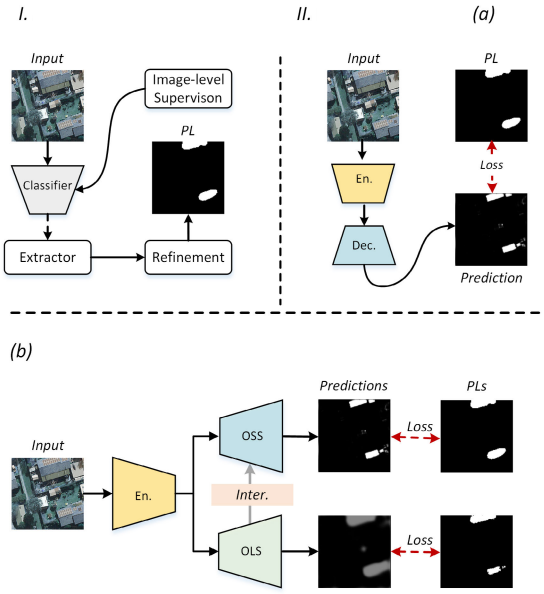


Fig. 1. Pipeline for weakly supervised object segmentation typically involves two key stages: an initialization stage for generating PLs and a refinement stage to produce more accurate predictions. (a) In vanilla frameworks, a single-path segmentation model is employed, which is often susceptible to the inductive biases introduced by single-source PLs. (b) In contrast, the proposed CJLF leverages both an OSS and an OLS, each supervised by PLs with complementary characteristics. The interaction between these streams allows the model to capture complementary information from the varied PLs, thereby mitigating the impact of inherent noise and inductive bias.

2) *Object Ambiguity Caused by Large Intraclass Variations and Low Contrast to Complex Urban Surroundings:* Remote sensing images, often limited in spatial resolution, depict solar panels with a wide range of sizes and shapes, depending on their installation settings. Due to the complicated urban environment, the target objects may share similar visual features with other objects. Such similarity causes ambiguity and confusion, thereby leading to false positives during the mapping process. It is noteworthy that the limited supervision provided by weak labels amplifies these challenges.

To learn from limited supervision, a widely studied weakly supervised object and semantic segmentation scheme involves a two-stage alternative training [12], [13], [14], as illustrated in Fig. 1. The initialization stage involves generating pixel-wise pseudo-labels (PLs) using a well-trained classifier with weak labels and an object localization extractor, followed by an optional refinement strategy to improve the PL quality. These PLs are then treated as ground truths in the refinement stage to train a segmentation model. The majority of current research efforts have concentrated on two main areas: improving the quality of noisy PLs [15] in the first stage and incorporating auxiliary guidance [16], [17] in the second stage. The quality of the PLs is crucial, as it directly impacts the effectiveness and accuracy of the segmentation model. Current weakly supervised object segmentation models rely on single-source PLs, which have inherent limitations due to the inductive biases introduced by the object localization extractors. The design of these extractors can influence how accurately they

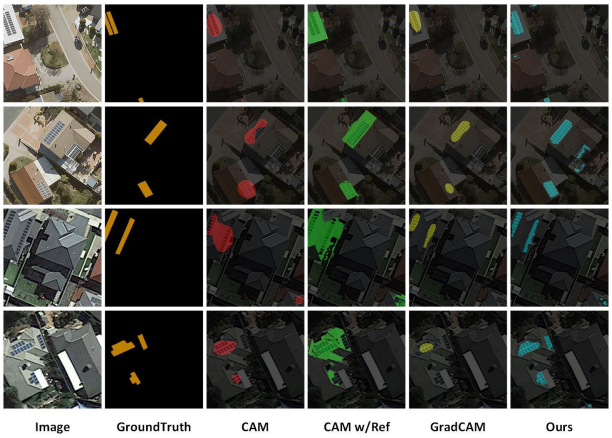


Fig. 2. Examples from CAM, CAM w/Ref, and GradCAM generated PL, demonstrating the inductive biases existing in the single source PLs and complementary characteristics of CAM w/Ref, and GradCAM.

distinguish the targets from the background, leading to the successful identification of some instances while misrepresenting or overlooking others. As illustrated in Fig. 2, PLs generated by class activation mapping (CAM) [18] often capture only a portion of the target objects. While gradient-weighted CAM (GradCAM) [19] improves precision, it still suffers from object boundary mismatches. CAM with conditional random field (CRF)-based refinement (CAM w/Ref) techniques can better identify target regions but struggle to differentiate co-occurring objects from the targets.

Rethinking the role of PLs, we propose that the complementary supervision provided by multiple types of PLs can be automatically integrated to enhance the model training, assuming that these PLs offer at least some complementary information. To this end, we propose a unified joint learning framework, the complementary joint learning framework (CJLF), to simultaneously capture the complementary supervision from diverse types of PLs. CJLF is built upon a shared encoder and dual-stream decoder architecture. The encoder leverages a residual-aggregation (RA)-based design to generate multiscale feature representations, which are crucial for capturing the varied and complex characteristics of input images. The dual-stream decoder consists of two specialized pathways: the object segmentation stream (OSS) and the object localization stream (OLS), supervised by two complementary kinds of PLs. The OSS is dedicated to producing fine-grained predictions with precise boundaries, ensuring accurate segmentation of target objects, while the OLS is designed for robust target localization. For OSS, we design a cross-scale attention accumulation strategy (CSAA)-based decoder to improve the segmentation accuracy, while a lightweight decoder is adopted for OLS. The attention fusion module (AFM) serves as a bridge between the OSS and OLS, allowing the two streams to complement each other. Coupled with the mask cross-entropy (MCE) loss, a subspace contrast constraint (SCC) is designed for model optimization by maximizing the feature discrepancy between the targets and the visually similar surroundings. The main contributions of this article are summarized as follows.

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1) A novel CJLF is proposed for weakly supervised SPM, establishing a new paradigm for effectively exploiting complementary supervision and combating PL noise.
2) To address the challenges of object ambiguity in complex backgrounds, we develop a cross-scale attention accumulation strategy, with AFM and Orthogonal Projection Operation (OPO) integrated into the OSS, ensuring the retention of fine-grained spatial details across scales.
3) We introduce a subspace contrast constraint to strengthen the model's ability to differentiate target objects from visually similar or co-occurring distractors against inherent PL noise.

Some of the concepts introduced in this article were preliminarily reported in [20]. Here, we present the complete and refined formulation, along with an extensive experimental evaluation of the proposed method. The remainder of the article is organized as follows. Section II reviews related work on weakly supervised semantic and object segmentation. Section III details the proposed CJLF. Section IV discusses experimental results and analysis. Finally, Section V concludes the article.

II. Related Work

A. General Weakly Supervised Semantic Segmentation

With the advantage of reducing annotation burdens, WSL has attracted increasing attention in the machine-learning community in recent years and has achieved remarkable performance in numerous tasks [21], [22], [23]. Different from the fully supervised learning framework, WSL aims to learn predictive models from training data with weak labels, which only offer partial and erroneous supervision information regarding the targets [21]. Inferring object localization is a crucial step to bridge the supervision gap in the WSL framework, leveraging weak annotations to generate coarse pixel-wise object localization, also known as PLs. The quality of PLs, however, is far from perfect [15]. They are inevitably involved with limited location accuracy and inconsistent quality in object coverage. To further extend the PLs to well-defined results, an alternative training scheme is proposed, which generally includes two learning stages: the initialization and the refinement stage (refer to Fig. 1).

The initialization step aims to identify regions of interest characterized by appearance variations and scale diversity and provide PLs that closely approximate the ground truth. Popular techniques include CAM [18], GradCAM [19], GradCAM++ [24], ScoreCAM [25], LayerCAM [26], and ToCo [27]. Refinement techniques, including multidirection filters [28], CRF [29], and semantic affinities [30], are introduced to further improve the initial PLs. Recently, vision and vision-language foundation models, such as the segment anything model [15], [31], and CLIP [32], have been incorporated to generate class-specific object masks.

The subsequent refinement stage endeavors to use the supervisory information in PLs for the training of accurate predictors for the target objects. Given the inherently noisy nature of PLs, the refinement stage should learn robust models to combat noisy supervision with inaccurate localization and incomplete target regions. The improvement of the refinement stage is typically brought by incorporating auxiliary guidance including saliency prior [17], edge information [16], and semantic affinities [33], developing consistency regularizer [34], [35], [36] and using training strategies [37] such as iterative training [38], [39], [40], and curriculum learning [41].

B. Weakly Supervised Object Segmentation in Remote Sensing Images

For remote sensing images, learning from weakly annotated samples for the object segmentation task can be more challenging. The research endeavor focuses on coping with the inherent characteristics of remote sensing images, including a sparser distribution of objects, ambiguous boundaries due to the limited spatial resolution, complicated background, and the imbalance between the foreground and background. Built on the two-step alternative training scheme [42], hierarchical feature representation [43], curriculum learning [44], iterative training [45], [46], attention mechanism [47], [48], [49], and uncertainty estimation [50], [51] have been actively studied to improve segmentation performance further. Recently, large pretrained models based on self-supervised learning, such as DINO [52], have been incorporated to suppress PL noise.

Recently, weakly supervised SPM has gained growing attention from the remote sensing community [2], [3], [51], but has not been extensively explored. Yu et al. [2] develop a weakly supervised method specially designed for SPM, in which Inception-v3 [53] and CAM are incorporated to obtain integral objects in completeness. Zhang et al. [3] propose a self-paced residual aggregated network, with the network architecture specially designed for preserving detailed object boundaries by residual feature aggregation. Zhang et al. [51] formulate the WS-SPM task as a label-noise learning problem and build a statistically consistent mapping model by estimating the instance-dependent transition matrix.

III. Methodology

A. Framework Overview

The basic idea of CJLF is to take advantage of the complementary supervision information of two PLs types, and thus help the model learn discriminative feature representation under the weakly supervised setting. To achieve this goal, CJLF has been developed with a shared encoder and dual-stream decoder architecture. The overview of the proposed method is presented in Fig. 3.

Given input image $x \in \mathbb{R}^{h\times w\times c}$, we have two noisy PL types, $y^p \in \mathbb{R}^{h\times w}, p = 1, 2$. The goal of CJLF is to learn an accurate predictive model $f(\theta)$ to assign each pixel $x_{i,j}$ a label $y_{i,j} \in \{0, 1\}$. To generate discriminative feature representations, a RA-based encoder is used to build multiscale feature embeddings $\{\mathcal{F}_l, l = 1, 2, \ldots, L\}$, where l denotes each level from low to high. The decoder of CJLF has two paths, OSS and OLS, supervised by two different types of PLs $y^p, p = 1, 2$. In our work, y^1 and y^2 are generated by GradCAM [19] and CAM w/Ref [28], respectively. The two types of PLs exhibit complementary characteristics: y^1 offers higher

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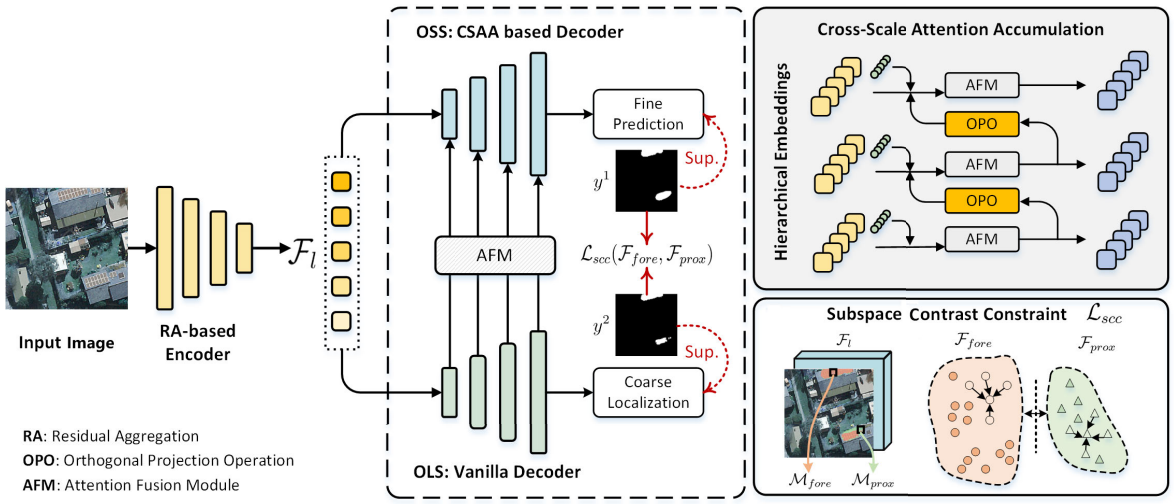


Fig. 3. Overall pipeline of the proposed method. The architecture of the CJLF comprises an RA-based encoder and a dual-stream decoder. The RA-based encoder focuses on generating discriminative feature embedding, whereas the dual-stream decoder operates with two specialized streams: the OSS and the OLS. In the training phase, both OSS and OLS will be optimized. Two different types of PLs, y^1 and y^2, serve as the supervision for the two streams, respectively. In the testing phase, only the OSS will be utilized for fine prediction.

precision, while y^2 achieves greater recall. More details about the PLs generation will be presented in the implementation details.

Sharing the same pyramidal features as inputs, the two parallel decoders in the CJLF are developed for different purposes: OSS aims to integrate hierarchical features $\{\mathcal{F}_l\}$ to produce accurate segmentation results. To this end, OSS is designed with a cross-scale attention accumulation strategy, combining the merits of AFM and OPO. By utilizing self-attention and cross-attention mechanisms, AFM aims to extract both local and global correspondence of the feature embedding $\{\mathcal{F}_l\}$ and identify complementary object localization clues from the features generated in the OLS. This establishes the interbranch interactions between the two streams, so the multiscale decoder features in OSS and OLS can complement each other. OPO accumulates the minor differences φ_l found by the AFM and integrates these differences from coarse to fine levels for full region representation. OLS is focused on precise object localization. Unlike OSS, OLS borrows the lightweight structure from the feature pyramid network (FPN) [54], enabling efficient localization of potential region candidates. Despite this, the simple decoder structure inevitably introduces ambiguous regions in predictions.

The loss function of CLJF consists of three parts: the mask cross-entropy loss $\mathcal{L}_{\text{mce}}$ for OSS, binary cross-entropy loss $\mathcal{L}_{\text{bce}}$ for OLS, and subspace contrast constraint $\mathcal{L}_{\text{scc}}$. We use $\mathcal{M}$ and $\mathcal{M}_d$ to denote the regions where PLs y^p agree and disagree, respectively. To reduce the model uncertainty in identifying $\mathcal{M}_d$, for the hierarchy l in the CSAA-based decoder, $\mathcal{L}_{\text{scc}}$ randomly selects pixels from the foreground mask in $\mathcal{M}$ and $\mathcal{M}_d$, pushes their foreground representation away from the distractors in the background and makes the intraclass feature more compact in the feature space. The objective for CLJF can be formulated as

$$\mathcal{L} = \mathcal{L}_{\text{mce}}\left(\tilde{y}, y^1\right) + \alpha \cdot \mathcal{L}_{\text{ce}}\left(\widehat{y}, y^2\right) + \beta \cdot \mathcal{L}_{\text{scc}} \tag{1}$$

where α is the parameter balancing the contributions from OSS and OLS. β is the parameter controlling the impact of SCC. During inference, the OSS prediction $\tilde{y}$ is retained as the final result, and the OLS output $\widehat{y}$ is discarded.

B. RA-Based Encoder

The RA-based encoder in CJLF consists of four residual aggregated blocks (RABs) for multiscale feature extraction. In $\{\mathcal{F}_l, l = 1, \ldots, L\}$, the feature map $\mathcal{F}_l$ has the size of $1/(2^{(i-1)})$ of the original input image. The first-level feature $\mathcal{F}_1$ is obtained by feeding the input image into two convolutional layers consecutively. The encoder has a top-down pathway, which consists of five stages, each comprising a 2×2 max-pooling operation for down-sampling, followed by a RAB:

$$\mathcal{F}_l = \text{MaxPool.}\left(\text{RAB}\left(\mathcal{F}_{l-1}\right)\right), i = 2, 3, \ldots, 6. \tag{2}$$

At each level, an additional 1×1 convolutional layer is leveraged to transform the feature maps into the same number of channels $d = 256$. Each RAB is composed of two sequential residual blocks (RBs) [55], followed by two convolutional layers using 3×3 kernels with a stride of 1 pixel. The output features from both RBs are concatenated and passed through the first convolutional layer. The resulting feature map is then processed by the second convolutional layer, which increases the number of channels by a factor of two. By propagating low-level residuals to the end of the block, RAB is designed to take advantage of multiscale residual features, thus helping predict complete objects with precise boundaries. The architecture of the RA-based encoder is detailed in [3].

C. Dual-Stream Decoder

The dual-stream decoder consists of two parallel paths: OSS and OLS, both utilizing a bottom-up architecture. Each encoder level is connected to the corresponding level in both OSS and OLS, ensuring effective integration of low-level

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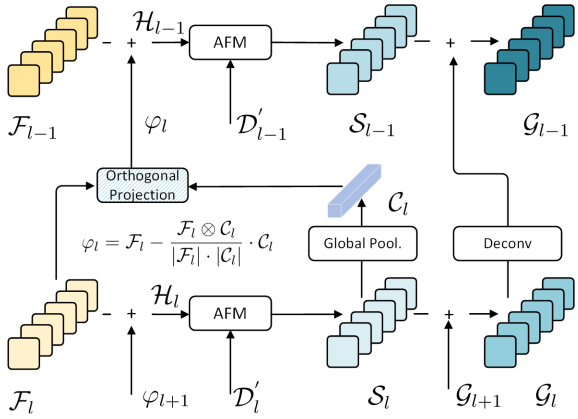


Fig. 4. Building block illustrating the orthogonal projection operation across the scales l and $l-1$.

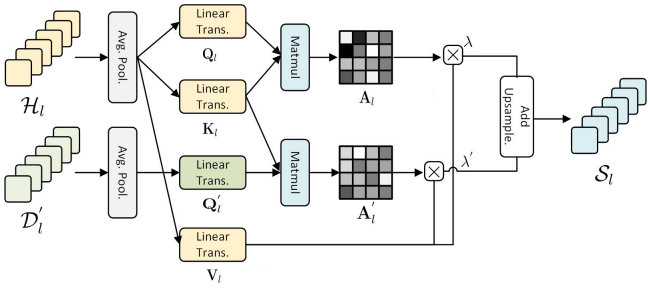


Fig. 5. Building block illustrating the structure of AFM at the lth level.

spatial details with high-level semantic features. The OSS path employs cross-scale attention accumulation, utilizing AFM and OPO to enhance performance. This complex structure also helps mitigate overfitting, particularly in WSL scenarios, where limited supervision can lead to a decline in segmentation accuracy and a loss of subtle spatial details, such as object boundaries. In contrast, the OLS path follows a simpler structure, duplicating the original FPN decoder design. In the CSAA-based OSS, the feature embedding at each hierarchy is fed into an AFM to capture both local and long-range relationships. Small differences will be separated from the original feature with an orthogonal projection and conveyed to the adjacent level for accurate localization. The general structure of CSAA is displayed in Fig. 4. The structure of CSAA will be introduced in more detail in Sections III-C1 and III-C2.

1) Attention Fusion Module: During the process of object identification within an image, human visual attention tends to focus on the main body of the object before shifting toward its subtle boundary, contrasting the background. Inspired by these observations, complementary learning is proposed between the two branches by taking advantage of both self- and cross-attention mechanisms. The multiscale decoder features from OLS are used as queries to compute the attention similarity scores with keys transformed from enhanced feature embedding, and then the features obtained by self- and cross-attention are adaptively fused. With AFM, the multiscale decoder features of OLS can provide object localization clues to OSS.

The structure of AFM is shown in Fig. 5. For each level l, we have the feature embedding $\mathcal{F}_l$ and discriminative feature discrepancy φ_{l+1} conveyed from the level $l+1$. The enhanced feature embedding, denoted as $\mathcal{H}_l \in \mathbb{R}^{h_l \times w_l \times c_l}$, is computed by adding the feature map φ_{l+1} to $\mathcal{F}_l$, scaled by a learnable weight w_{l+1}

$$\mathcal{H}_l = \mathcal{F}_l + w_{l+1} \times \varphi_{l+1}. \tag{3}$$

After the 2×2 averaging pooling operation, $\mathcal{H}_l$ is then fed into three convolutional layers $\text{Conv}_{l,k}, k = 1, 2, 3$, reducing the number of channels to $c_l/8$ to avoid heavy computational burdens and then reshaped into three vectors $\mathbf{Q}_l, \mathbf{K}_l, \mathbf{V}_l \in \mathbb{R}^{(h_l \times w_l) \times c_l}$

$$\mathbf{Q}_l, \mathbf{K}_l, \mathbf{V}_l = \text{reshape}\left[\text{Conv}_{l,k}\left(\text{Avg.Pool.}\left[\mathcal{H}_l\right]\right)\right], k = 1, 2, 3 \tag{4}$$

where reshape$[\cdot]$ and Avg.Pool.$[\cdot]$ denote the reshape and the average pooling operations, respectively.

Similarly, the multiscale decoder features of OLS $\mathcal{D}'_l \in \mathbb{R}^{h_l \times w_l \times c_l}$ will be linearly transformed into $\mathbf{Q}'_l$. Then, the self- and cross-attention similarity matrix can be computed by

$$\mathbf{A}_l\left(\mathbf{Q}_l, \mathbf{K}_l, \mathbf{V}_l\right) = \text{softmax}\left(\frac{\mathbf{Q}_l \mathbf{K}_l^T}{\sqrt{d_{k,l}}}\right)\mathbf{V}_l \tag{5}$$

$$\mathbf{A}'_l\left(\mathbf{Q}'_l, \mathbf{K}_l, \mathbf{V}_l\right) = \text{softmax}\left(\frac{\mathbf{Q}'_l \mathbf{K}_l^T}{\sqrt{d_{k,l}}}\right)\mathbf{V}_l \tag{6}$$

where $d_{k,l}$ is the dim of $\mathbf{Q}_l$ and $\mathbf{K}_l$. The attention enhanced feature representation $\mathcal{S}_l$ can be computed by the weighted sum

$$\mathcal{S}_l = \lambda_l \cdot \mathbf{A}_l \cdot \mathbf{V}_l + \lambda'_l \cdot \mathbf{A}'_l \cdot \mathbf{V}_l \tag{7}$$

where λ_l and λ'_l are learnable parameters.

2) Orthogonal Projection Operation: To identify the discriminative properties φ_l, which attract greater attention from AFM, the orthogonal projection operation is utilized to disentangle them from the original feature embedding $\mathcal{F}_l$. The global feature representation at the lth level $\mathcal{C}_l$ is computed by applying a global average pooling operation to the attention-enhanced feature $\mathcal{S}_l$ over the spatial dimensions

$$\mathcal{C}_l = \text{GlobalPool.}\left(\mathcal{S}_l\right). \tag{8}$$

Then, the discriminative feature at lth level φ_l can be obtained after subtracting the redundant common properties from $\mathcal{F}_l$

$$\varphi_l = \mathcal{F}_l - \frac{\mathcal{F}_l \otimes \mathcal{C}_l}{|\mathcal{F}_l| \cdot |\mathcal{C}_l|} \cdot \mathcal{C}_l \tag{9}$$

where $\otimes$ denotes the matrix multiplication.

The separated discriminative feature φ_l will be conveyed to the next adjacent level and integrated with original feature embeddings $\mathcal{F}_{l-1}$, with the learnable parameter ω_l used to adjust the impacts of φ_l. So, $\mathcal{G}_{l-1}$ can be obtained

$$\mathcal{S}_{l-1} = \text{AFM}\left(\mathcal{H}_{l-1}, \mathcal{D}'_{l-1}\right) \tag{10}$$

$$\mathcal{G}_{l-1} = \mathcal{S}_{l-1} + \text{Deconv}\left(\mathcal{G}_l\right). \tag{11}$$

To avoid a high computation burden, AFM is not applied to the first level

$$\mathcal{G}_1 = \mathcal{F}_1 + \text{Deconv}\left(\mathcal{G}_2\right). \tag{12}$$

Finally, three consecutive convolutional layers and soft-max activation are added to produce the fine prediction $\tilde{y}$.

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D. Subspace Contrast Constraint

In typical WSL training procedures, unlabeled or uncertain pixels are often excluded from loss computation because they are believed to lack useful information and may even introduce erroneous supervision. However, if assumptions can be made about the classes these pixels belong to, they can provide valuable information for the training process. Hence, we introduce a subspace contrast constraint, which enforces feature space constraints to enhance the model's discriminative ability in these ambiguous regions, particularly in distinguishing target objects from co-occurring objects in the surroundings. To this end, we propose maximizing the distance between interclass feature representations while minimizing the distance between pixels within the same class, thereby promoting compact and class-consistent representations.

In our design, mask cross-entropy loss $\mathcal{L}_{\text{mce}}$ is utilized to calculate the loss for the pixels that have consistent labels, where $\mathcal{M}$ denotes the mask for pixels with the same labels. The exclusive disjunction operation is implemented to obtain the mask $\mathcal{M}_d$ for ambiguous regions from y^1 and y^2. Parameter γ is then used to reduce the impact of these pixels. $\mathcal{L}_{\text{mce}}$ can be computed as follows:

$$\mathcal{L}_{\text{mce}} = \frac{1}{N}\left(\sum_{i,j\in\mathcal{M}} \mathcal{L}_{\text{ce}}\left(y_{i,j}, y^1_{i,j}\right) + \gamma \cdot \sum_{i,j\in\mathcal{M}_d} \mathcal{L}_{\text{ce}}\left(y_{i,j}, y^1_{i,j}\right)\right) \tag{13}$$

where i, j denotes the pixel location. N is the total number of pixels.

To further improve the model's ability to identify foreground pixels from the background in uncertain regions, $\mathcal{L}_{\text{scc}}$ is calculated by comparing intraclass and interclass feature similarities. Considering the potential errors in PLs, we randomly select K pixels from the foreground mask $\mathcal{M}$, and pixels with inconsistent PLs in $\mathcal{M}_d$, which are denoted as $\mathcal{M}^*_{\text{fore}}$ and $\mathcal{M}^*_d$, respectively. The inter and intraclass cosine similarities of these pixels are calculated as follows:

$$\psi_l = \sum_{m,n\in\mathcal{M}^*_{\text{fore}}} \mathbf{1}_{[m\neq n]} \exp\left(\text{sim}\left(\mathcal{F}_{l,m}, \mathcal{F}_{l,n}\right)/\tau\right) \tag{14}$$

$$\psi'_l = \sum_{m,n\in\mathcal{M}^*_d} \mathbf{1}_{[m\neq n]} \exp\left(\text{sim}\left(\mathcal{F}_{l,m}, \mathcal{F}_{l,n}\right)/\tau'\right) \tag{15}$$

where sim denotes the cosine similarity. $\mathcal{F}_{l,m}$ and $\mathcal{F}_{l,n}$ represent the mth and nth vectors in the feature embedding at the lth level. $\mathbf{1}_{[m\neq n]} \in \{0, 1\}$ is the indicator function. τ and τ' denote the temperature parameters for the two classes, respectively. The interclass similarity is calculated by

$$\psi^*_l = \sum_{m\in\mathcal{M}^*_{\text{fore}}, n\in\mathcal{M}^*_d} \exp\left(\text{sim}\left(\mathcal{F}_{l,m}, \mathcal{F}_{l,n}\right)/\tau^*\right) \tag{16}$$

where τ^* denotes the temperature parameter balancing the contribution of the interclass similarity. To achieve the goal of increasing intraclass compactness and interclass separation, the hierarchical subspace contrast loss is defined as

$$\mathcal{L}_{\text{scc}} = -\log\frac{\psi_l + \psi'_l}{\psi^*_l}. \tag{17}$$

To avoid the over-fitting issue and simultaneously keep subtle boundary details, $l = 2$ is used to calculate the $\mathcal{L}_{\text{scc}}$. Greater intraclass feature similarity and smaller interclass similarity result in a lower value of $\mathcal{L}_{\text{scc}}$.

IV. Experiments and Discussion

A. Experiments Setup

1) Datasets: In the experiments, the performance of the proposed method is validated on three challenging datasets for solar panel mapping, including two aerial-image datasets, AerialGDA2020 and DualGMS, and one satellite-image dataset, GloSoFarID.

AerialGDA2020 was collected from the aerial imagery provided by the Geocentric Datum of Australia 2020, identifying more than 20 000 solar panels in Australian Capital Territory (ACT), Australia. The original size of the images is $10\,000 \times 10\,000$ pixels covering 1 km^2, with 75-mm spatial resolution and RGB bands. The original images were cropped into patches with the size of 512×512 pixels. The training set of AerialGDA2020 consists of 10 988 positive samples and 10 309 negative samples, while the test set contains 2673 positive samples. The validation set with 80 image patches is used to monitor the model fit on the training dataset. For the test set, experts in remote sensing drew polygons of solar panels manually, and these polygons are used to generate binary pixel-wise masks as the ground truth.

DualGMS was collected from Google Static Maps with the aerial images captured over Canberra, ACT, and Brisbane, Queensland, Australia. The training set of DualGMS consists of 1588 positive samples and 1800 negative samples. The test set has 99 positive samples. All the images in the DualGMS datasets have a size of 256×256 with spatial resolution varying from 15 to 30 cm.

GloSoFarID [57] comprises medium-resolution Sentinel-2 multispectral imagery (13 bands) focused on both large-scale and small-scale solar farms. Its updated 2021–2023 data offers a contemporary and expanded view of the global solar landscape. GloSoFarID includes both positive samples (containing solar farms) with pixel-level annotations and negative samples (without solar farms), making it suitable for training classification models and generating PLs for segmentation tasks. In our experiment, we utilized a subset of the dataset consisting of 888 positive samples and 787 negative samples for training, along with a test set of 157 positive samples, using RGB bands. Compared to the two aerial datasets, GloSoFarID presents a more challenging setting: it is characterized by greater scene heterogeneity, much lower spatial resolution, and highly complex backgrounds.

2) Evaluation Metrics: For the quantitative evaluation of the proposed method, seven widely adopted metrics for foreground object detection are employed, i.e., Accuracy (AC), Precision (P), Recall (R), F measure, mean intersection over union (mIoU), mean square error (mse), and enhanced-alignment measure (E) [16]. In our work, the mean F measure, mean IoU, and E measure are calculated over the test sets. The F-measure curves and PR curves are also utilized to reveal the method's performance versus varying threshold.

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TABLE I

QUANTITATIVE EVALUATION RESULTS ON THE AERIALGDA2020 AND DUALGMS DATASETS. THE BEST RESULTS ARE IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED. ↑ AND ↓ INDICATE THAT LARGER AND SMALLER RESULTS ARE BETTER, RESPECTIVELY. MEAN F_1 AND MEAN IOU ARE ADOPTED

		AerialGDA2020							**DualGMS**						
Method	Venue	$AC\uparrow$	$P\uparrow$	$R\uparrow$	$F_1\uparrow$	$mIoU\uparrow$	$MSE\downarrow$	$E\uparrow$	$AC\uparrow$	$P\uparrow$	$R\uparrow$	$F_1\uparrow$	$mIoU\uparrow$	$MSE\downarrow$	$E\uparrow$
CAM [18]	CVPR 2016	0.9848	0.6568	0.6519	0.6028	0.4515	0.0257	0.8334	0.9639	0.5604	0.5383	0.5239	0.4000	0.0537	0.7680
GradCAM [19]	ICCV 2017	0.9885	0.7102	0.7462	0.7096	0.5662	0.0304	0.9084	0.9717	0.9022	0.4395	0.5728	0.4141	0.0433	0.7490
ACoL [56]	CVPR 2018	0.8515	0.1577	0.9286	0.2597	0.1557	0.1788	0.4272	0.8849	0.2593	**0.9172**	0.3859	0.2470	0.1407	0.5378
WS-SOD [16]	CVPR 2020	0.9874	0.6336	0.9234	0.7356	0.5980	0.0348	0.8768	0.9687	0.5967	0.7452	0.6321	0.4958	0.0328	0.8339
LayerCAM [26]	TIP 2021	0.9681	0.3484	0.8979	0.4854	0.3293	0.0497	0.6760	0.9578	0.7282	0.3564	0.4176	0.2799	0.0807	0.6757
MFNet [28]	ICCV 2021	0.9791	0.5219	0.8844	0.6256	0.4780	0.0219	0.7974	0.9703	0.7171	0.5116	0.5521	0.4151	0.0310	0.8017
SCWSSOD [35]	AAAI 2021	0.9889	0.7704	0.6516	0.6800	0.5376	**0.0119**	0.8748	0.9652	0.8127	0.2007	0.2993	0.1958	0.0348	0.4727
NSAL [13]	TMM 2022	0.9631	0.3371	**0.9785**	0.4928	0.3337	0.0375	0.6317	0.9768	0.7708	0.5092	0.5573	0.4435	0.0261	0.7303
UASR [48]	TGRS 2022	0.9862	0.5942	0.9170	0.7059	0.5545	0.2765	0.8581	0.9675	0.8442	0.3943	0.5153	0.3845	0.2843	0.6782
SP-RAN [3]	TGRS 2022	0.9890	0.7053	0.8751	0.7641	0.6344	0.0204	0.9084	0.9816	**0.9365**	0.6178	0.7271	0.5937	0.0380	0.8842
SCW [37]	PRCV 2023	0.9855	0.5890	0.8830	0.6876	0.5411	0.0231	0.8501	0.9697	0.7126	0.4118	0.4768	0.3653	0.0306	0.6688
WeSAM [14]	CVPR 2024	0.9840	0.6611	0.2867	0.3518	0.2790	0.0638	0.5298	0.9787	0.8717	0.6683	0.7287	0.6299	0.0400	0.8573
PLNM [40]	TNNLS 2025	0.9895	0.7209	0.7903	0.7402	0.5974	0.0122	**0.9262**	0.9781	0.8727	0.5827	0.6849	0.5346	0.0269	0.8664
CJLF		**0.9929**	**0.8076**	0.8346	**0.8049**	**0.6882**	0.0137	0.9252	**0.9851**	0.8766	0.6916	**0.7525**	**0.6378**	**0.0222**	**0.8947**

B. Implementation Details

The implementation of the proposed method consists of two steps: PLs generation and training of the object segmentation model. Both steps are implemented with Pytorch on a single NVIDIA V100 GPU. For the generation of PLs, GradCAM, and CAM w/Ref are adopted to obtain two different types of PLs with complementary characteristics. The backbones for GradCAM and CAM are VGG16 [58] and DenseNet-169 [59], both initialized by pretrained weights on ImageNet. For GradCAM, Adam optimizer is used to train the backbone with a batch size of 32. The initial learning rate is set as 5×10^{-3}, and a 0.9 decay is applied for every 2 epochs. The training process for all three datasets ends after 50 epochs. For CAM, Adam optimizer is used to train the backbones with batch sizes 6, 8, and 8 for AerialGDA2020, DualGMS, and GloSoFarID datasets, respectively. The initial learning rate is 10^{-4}, decayed by 0.9 for every epoch. The training process terminates after 50 epochs for all three datasets. The inferred activation maps are then refined by CRF with parameters $\theta_\alpha = 60$, $\theta_\beta = 7$, and $\theta_\gamma = 3$ on AerialGDA2020 and DualGMS datasets. For GloSoFarID, we set $\theta_\alpha = 60$, $\theta_\beta = 5$, and $\theta_\gamma = 3$.

During the training process of the mapping network, the Adam optimizer is used with a batch size of 8. Initial learning rate 1×10^{-3} is set for all datasets. The learning rate drops by 10% every 3, 6, and 3 epochs. For the loss in (1), hyperparameters $\alpha = 0.3$, $\beta = 10^{-3}, \gamma = 0.1$, $\alpha = 0.5$, $\beta = 5 \times 10^{-4}, \gamma = 0$, and $\alpha = 0.9$, $\beta = 1 \times 10^{-4}, \gamma = 0$ are set for AerialGDA2020, DualGMS and GloSoFarID datasets. For parameter K in (16), we set $K = 1000, 500, 900$ for AerialGDA2020, DualGMS and GloSoFarID datasets, separately.

The source code for CJLF will be publicly released at https://github.com/zhangjue1993/CJLF upon publication.

C. Comparison With the State of the Art

1) Methods for Comparison: To evaluate the performance of the proposed method, CJLF is compared with 13 state-of-the-art weakly supervised methods. They are CAM [18], GradCAM [19], LayerCAM [26], ACoL [56], WS-SOD [16], MFNet [28], SCW [37], SCWSSOD [35], UASR [48], SP-RAN [3], NSAL [13], WeSAM [14], and PLNM [40]. Among these methods, CAM, GradCAM, LayerCAM, and ACoL are weakly supervised object localization methods with image-level labels; MFNet, SCW, and NSAL are weakly supervised salient object detection methods proposed recently; WS-SOD and SCWSSOD are developed for weakly supervised salient object detection with scribble annotations; UASR is originally proposed for weakly supervised region of interest extraction with remote sensing images; SP-RAN are specially designed for WS-SPM in high-resolution aerial images. WeSAM is built on SAM with the bounding box annotations as the prompt.

TABLE II

QUANTITATIVE EVALUATION RESULTS ON THE GLOSOFARID DATASET. THE BEST RESULTS ARE IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED. ↑ AND ↓ INDICATE THAT LARGER AND SMALLER RESULTS ARE BETTER, RESPECTIVELY. MEAN F_1 AND MEAN IOU ARE ADOPTED

Method	Venue	$AC\uparrow$	$P\uparrow$	$R\uparrow$	$F_1\uparrow$	$mIoU\uparrow$	$MSE\downarrow$	$E\uparrow$
GradCAM [19]	ICCV 2017	0.8321	0.6304	0.4151	0.4188	0.2915	0.2178	0.6430
MFNet [28]	ICCV 2021	0.8416	0.7158	0.5410	0.5398	0.4422	0.1590	0.6791
UASR [48]	TGRS 2022	0.8754	**0.8176**	0.3732	0.4790	0.3603	0.3428	0.6224
SP-RAN [3]	TGRS 2022	**0.9063**	0.7326	**0.6370**	0.6388	0.5193	**0.1204**	0.7970
WeSAM [14]	CVPR 2024	0.8657	0.6762	0.6112	0.6092	0.5008	0.1452	0.7557
PLNM [40]	TNNLS 2025	0.9017	0.7329	0.6211	0.6366	0.5083	0.1433	**0.8057**
CJLF		0.9051	0.7885	0.6356	**0.6627**	**0.5395**	0.1305	0.8023

2) Quantitative Evaluation: Tables I and II present the quantitative performance comparison between the proposed method and the state-of-the-art approaches. The best results are highlighted in bold, while the second-best results are underlined. The symbols ↑ and ↓ indicate whether higher or lower values are better, respectively. Both CAM and GradCAM demonstrate consistent and stable performance across the AerialGD2020 and DualGMS datasets. However, the modified WSOL methods, such as LayerCAM and ACoL, exhibit limitations in remote sensing scenarios, particularly in balancing precision and recall. ACoL, while highly sensitive (with high recall), suffers from an excessive number of false positives due to low precision, making it less reliable for

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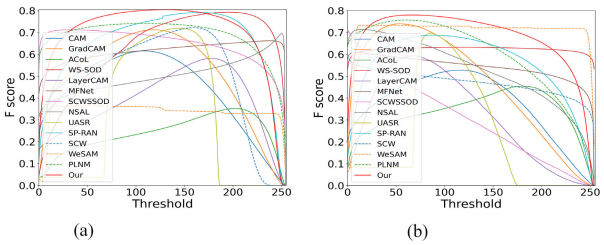


Fig. 6. F-measure curves for the state-of-the-art methods and the proposed CJLF on (a) AerialGDA2020 and (b) DualGMS datasets.

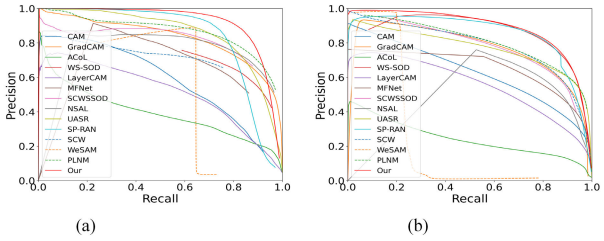


Fig. 7. Precision–recall curves for the state-of-the-art methods and the proposed CJLF on (a) AerialGDA2020 and (b) DualGMS datasets.

the WS-SPM task. On the other hand, MFNet and SCW, utilizing the alternative training scheme, consistently outperform LayerCAM and ACoL on both the AerialGDA2020 and DualGMS datasets, achieving significant improvements in both F_1 and mIoU scores. WS-SOD, leveraging scribble annotations, excels in precision, highlighting its effectiveness in accurately detecting objects. Incorporating the vision foundation model SAM, WeSAM achieves a good balance between precision and recall on the DualGMS dataset, but struggles to perform well on AerialGDA2020. Notably, these methods were originally designed for natural scene segmentation. For remote sensing-specific tasks, UASR shows improved capability in managing complex backgrounds, although it still struggles with differentiating foreground objects from the background. SP-RAN, optimized for high-resolution aerial imagery, delivers notable improvements, with an approximate 6% and 7% increases in F_1 and mIoU scores, respectively, compared to GradCAM. Most significantly, the proposed CJLF method not only demonstrates stable and consistent performance across both datasets but also shows substantial gains in quantitative metrics. Compared with SP-RAN, 4% and 5.4% increases on F_1 and mIoU scores on the AerialGDA2020 dataset, and 2.5% and 4.4% increases on F_1 and mIoU scores on DualGMS. On the GloSoFarID dataset, as shown in Table II, the proposed CJLF achieves the best overall performance, with the highest mean F_1 (0.6627) and mIoU (0.5395). CJLF outperforms the strongest prior baseline SP-RAN by 2.39%in F_1 and 2.02% in mIoU, and also surpasses recent methods such as PLNM (+ 2.61% F_1, + 3.12% mIoU) and WeSAM (+ 5.35% F_1, + 3.87% mIoU). Notably, CJLF maintains a strong balance between precision and recall while remaining highly competitive on the remaining criteria.

To further validate the quantitative performance of CJLF, we show the F-measure curves and PR curves in Figs. 6–8. In the PR curves, methods with curves closer to the top right corner

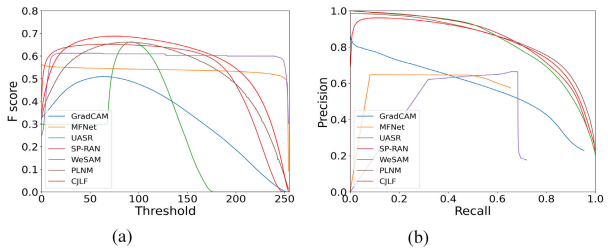


Fig. 8. F-measure curves and Precision–recall curves for the state-of-the-art methods and the proposed CJLF on GloSoFarID dataset. (a) F-score. (b) PR curves.

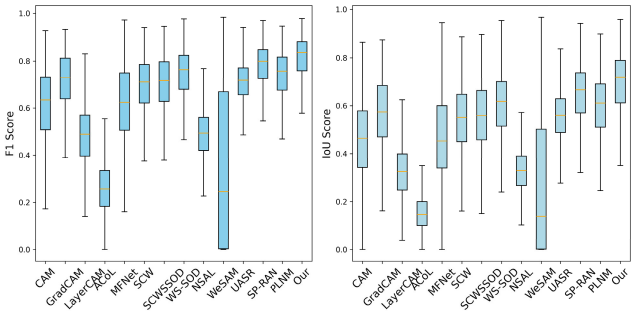


Fig. 9. Box plots showing the distribution of F_1 and IoU scores for methods on the AerialGDA2020 dataset, with the F_1 scores on the left and IoU scores on the right.

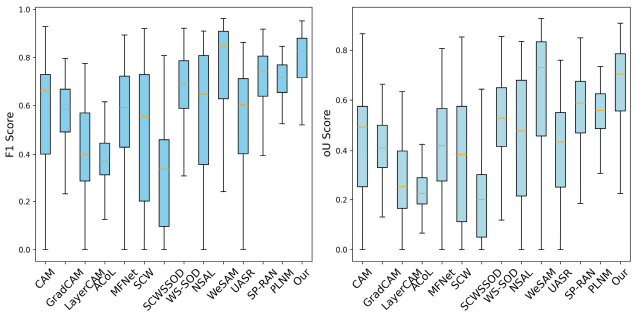


Fig. 10. Box plots showing the distribution of F_1 and IoU scores for methods on the DualGMS dataset, with the F_1 scores on the left and IoU scores on the right.

demonstrate superior performance, while in the F curves, methods with curves closer to the top demonstrate higher F_1 across thresholds. In all figures, red lines are used to denote the proposed CJLF. As shown in the figures, CJLF provides consistently strong F-measure performance across thresholds and a favorable precision–recall tradeoff.

Figs. 9–11 present box plots that summarize the per-image performance distributions of different methods on the three datasets, in terms of F_1 and IoU scores. Each method is represented on the x-axis, and the y-axis shows the respective metric score ranging from 0 to 1. The box plots give a visual representation of the distribution and variability of the quantitative performance of the proposed CJLF and the methods for comparison. Boxes represent the interquartile range (IQR), which covers the middle 50% of the data. Whiskers extend from the boxes to 1.5 times the IQR. The

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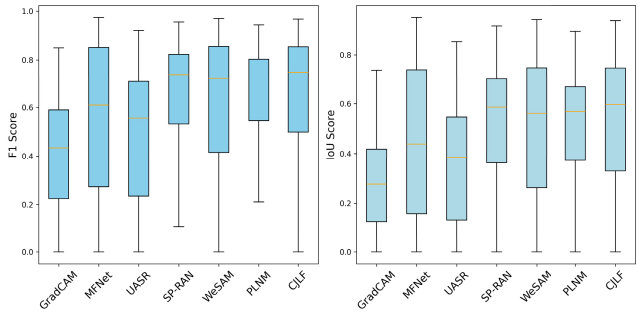


Fig. 11. Box plots showing the distribution of F_1 and IoU scores for methods on the GloSoFarID dataset, with the F_1 scores on the left and IoU scores on the right.

orange line inside each box indicates the median score for that method. Overall, the box plots provide an intuitive view of both central tendency and variability. MFNet exhibits substantial dispersion, suggesting unstable performance across images, whereas LayerCAM and ACoL show consistently lower medians, indicating inferior segmentation quality. In contrast, SP-RAN, PLNM, and the proposed CJLF achieve higher medians with relatively compact boxes, demonstrating more stable and robust performance. Notably, CJLF maintains strong central performance while limiting variance, implying that its gains are consistent across diverse images rather than driven by a few favorable cases.

3) Qualitative Evaluation: To further demonstrate the effectiveness of the proposed CJLF, representative examples are shown for visual comparison in Figs. 12–14. The results generated by different methods exhibit clear differences. CAM identifies the general object regions but struggles with precise localization, often highlighting only partial region candidates. GradCAM offers improved identification of discriminative object parts but still falls short in complete object coverage. LayerCAM delivers inconsistent performance across datasets, with noticeable variations in precision and recall. The use of adversarial learning in ACoL enhances object completeness, yet it struggles to distinguish objects from co-occurring regions, leading to inferior performance. MFNet, utilizing CRF for refinement, achieves a good balance between object completeness and boundary clarity. SCW, though showing varied performance, offers improved boundary preservation and localization, particularly in the AerialGDA2020 dataset, but is prone to missing objects in DualGMS. SCWSSOD exhibits poor performance in discovering objects, missing significant instances. NSAL is capable of accurately locating instances but fails to distinguish them from adjacent objects. WS-SOD maintains a solid balance between precision and recall but lacks the ability to differentiate between targets and co-occurrences effectively. UASR and SP-RAN improve overall detection, but their predictions are frequently over-smoothed and prone to false positives, particularly around boundaries and over textured or nonuniform backgrounds. WeSAM preserves object contours more effectively; however, it may miss small-scale instances and sometimes produces scattered activations and slightly misaligned boundaries in challenging cases on the DualGMS and GloSoFarID datasets. On both DualGMS and GloSoFarID datasets, while PLNM delivers accurate localization with relatively clean masks, its boundaries remain suboptimal, with blurred or slightly misaligned contours near object edges. On the AerialGDA2020 dataset, PLNM yielded incomplete predictions (see the third and fourth rows in Fig. 12). In contrast, the proposed CJLF method delivers superior performance, consistently producing results closer to the ground truth. It excels in both object completeness and boundary accuracy and is robust in handling objects of varying sizes and shapes against complex backgrounds. The method shows the most reliable balance of precision, recall, and boundary detection, making it suitable for complex remote sensing scenes.

D. Ablation Study

1) Impact of the Components of CJLF: Ablation studies are conducted to reveal the effectiveness and impacts of each component in CJLF, i.e., AFM, OPO, OLS, and SCC. We systematically remove each component from the full CJLF model and observe the corresponding performance changes. M0 denotes the baseline of our work, consisting of an RA-based encoder and a basic FPN decoder with a self-attention module applied to the same level. M0 is supervised by PLs from GradCAM with binary cross-entropy. M1 adds OPO to M0. In Table III, method M2–M4 represents different configurations of the CJLF, where components are selectively removed, while ours refers to the complete version of CJLF with all components enabled. The quantitative results are reported in Table III while the visual comparison is shown in Figs. 15 and 16. Compared with the baseline M0, M1–M4 present varied improvement.

Effect of AFM: The importance of AFM can be revealed by comparing the quantitative results of M4 and CJLF, with the F_1 and mIoU scores increasing by around 1% and 1.5% on the AerialGDA2020 dataset and 8% and 9.4% on DualGMS.

Effect of OPO: OPO contributes to the accumulation of the minor but significant features discovered by the AFM, bringing 4.6% and 5.3% increases on the F_1 and mIoU scores for the DualGMS data and 1% and 1.3% increases on the F_1 and E scores on the AerialGDA2020 dataset.

Effect of OLS and SCC: OLS is proposed to provide complementary location cues for OSS and improve the completeness of the detected objects. SCC is designed to keep precise boundaries and simultaneously avoid the side-effect of complementary learning: over-predictions. As both AFM and SCC are intrinsically dependent on the OLS branch, removing OLS simultaneously disables both modules. Thus, to quantify the impact of excluding the OLS, we report this configuration as M1 (w/o OLS, AFM, and SCC). The contribution of the SCC is assessed by comparing M3 (w/o SCC) with CJLF. We can observe that the employment of SCC achieves 2.6% and 2.8% increases on the F_1 and mIoU scores on AerialGDA2020 and 0.6% and 1.1% increases on the DualGMS dataset.

Figs. 15 and 16 visually illustrate the performance differences between the proposed CJLF and its degraded variants. Both M0 and M1 exhibit limited abilities in locating targets

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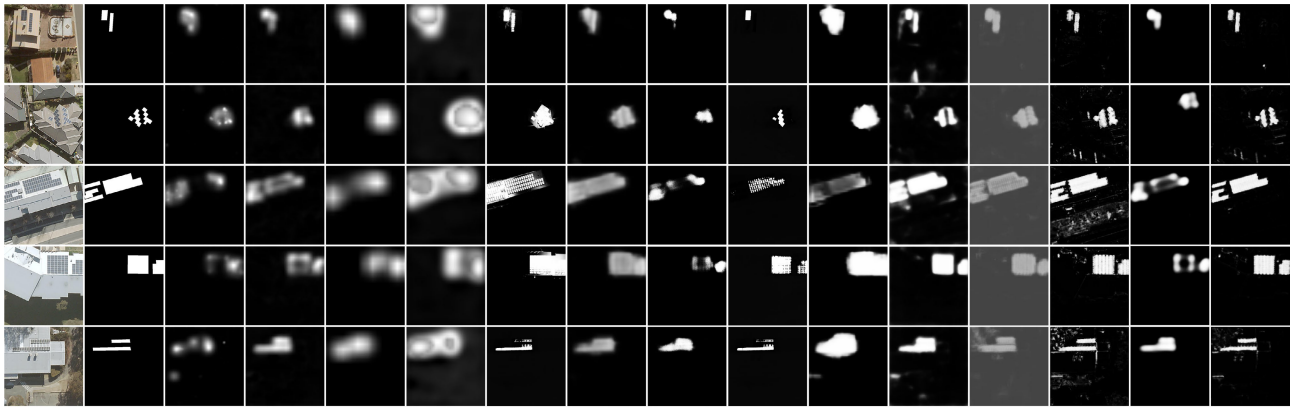


Fig. 12. Predicted mapping results of competing methods and our method on the AerialGDA2020 dataset. (a) Test images. (b) GT. (c) CAM. (d) GradCAM. (e) LayerCAM. (f) ACoL. (g) MFNet. (h) SCW. (i) SCWSSOD. (j) WeSAM. (k) NSAL. (l) WS-SOD. (m) UASR. (n) SP-RAN. (o) PLNM. (p) CJLF (ours).

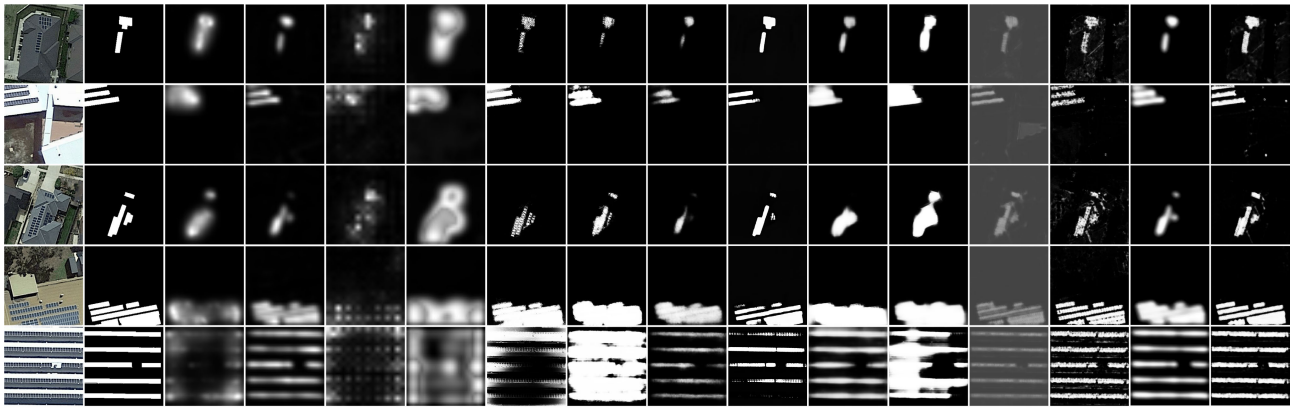


Fig. 13. Predicted mapping results of competing methods and our method on the DualGMS dataset. (a) Test images. (b) GT. (c) CAM. (d) GradCAM. (e) LayerCAM. (f) ACoL. (g) MFNet. (h) SCW. (i) SCWSSOD. (j) WeSAM. (k) NSAL. (l) WS-SOD. (m) UASR. (n) SP-RAN. (o) PLNM. (p) CJLF (ours).

TABLE III
QUANTITATIVE ABLATION STUDY RESULTS ON AERIALGDA2020 AND DUALGMS DATASETS

Data set	**Method**	**AFM**	**OPO**	**OLS**	**SCC**	AC	P	R	F_1	$mIoU$	MSE	E
AerialGDA2020	M0					0.9879	0.6728	0.8830	0.7457	0.6102	0.0256	0.8956
	M1		✓			0.9881	0.6791	**0.8855**	0.7509	0.6172	0.0249	0.8970
	M2	✓		✓	✓	0.9914	0.8172	0.8127	0.7942	0.6819	0.0178	0.9126
	M3	✓	✓	✓		0.9921	**0.8296**	0.7734	0.7782	0.6609	0.0137	0.9036
	M4		✓	✓	✓	0.9924	0.7906	0.8317	0.7942	0.6731	0.0144	0.9235
	CJLF (Ours)	✓	✓	✓	✓	**0.9929**	0.8076	0.8346	**0.8049**	**0.6882**	**0.0137**	**0.9252**
DualGMS	M0					0.9797	0.8899	0.5300	0.6395	0.5139	0.0282	0.7846
	M1		✓			0.9798	0.8974	0.5267	0.6367	0.5097	0.0285	0.7853
	M2	✓		✓	✓	0.9833	**0.9057**	0.6134	0.7063	0.5846	0.0252	0.8505
	M3	✓	✓	✓		0.9849	0.8874	0.6739	0.7460	0.6264	0.0235	0.8915
	M4		✓	✓	✓	0.9812	0.8925	0.5661	0.6723	0.5436	0.0283	0.8227
	CJLF (Ours)	✓	✓	✓	✓	**0.9851**	0.8766	**0.6916**	**0.7525**	**0.6378**	**0.0222**	**0.8947**

and differentiating objects from nearby elements. In contrast, CJLF demonstrates a superior ability to identify a higher number of foreground pixels, particularly near object boundaries, when compared to M2 and M4. In addition, compared with M3, CJLF further minimizes ambiguity between objects and co-occurring elements, as well as interference from the background.

2) Impact of the Joint Supervision of PLs: To validate the effectiveness of the complementary joint learning paradigm, we utilize single-source PLs, i.e., GradCAM and CAM w/Ref,

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TABLE IV
QUANTITATIVE RESULTS USING DIFFERENT TYPES OF PLS ON AERIALGDA2020 AND DUALGMS DATASETS

Data set	PL Type	w M1	AC	P	R	F_1	$mIoU$	MSE	E
AerialGDA2020	GradCAM	✗	0.9885	0.7102	0.7462	0.7096	0.5662	0.0304	0.9084
		✓	0.9881	0.6791	**0.8856**	0.7509	0.6172	0.0249	0.8970
	CAM w/Ref	✗	0.9808	0.5660	0.8425	0.6465	0.4963	0.0193	0.8345
		✓	0.9879	**0.9220**	0.4924	0.5979	0.4720	0.0352	0.7246
	CJLF (Ours)	-	**0.9929**	0.8076	0.8346	**0.8049**	**0.6882**	**0.0137**	**0.9252**
DualGMS	GradCAM	✗	0.9717	0.9022	0.4395	0.5728	0.4141	0.0433	0.7490
		✓	0.9798	**0.9875**	0.5267	0.6367	0.5097	0.0285	0.7853
	CAM w/Ref	✗	0.7850	0.5030	**0.7345**	0.4886	0.3591	0.1860	0.7130
		✓	0.9561	0.4741	0.1170	0.1636	0.1026	0.2573	0.4920
	CJLF (Ours)	-	**0.9851**	0.8766	0.6916	**0.7525**	**0.6378**	**0.0222**	**0.8947**

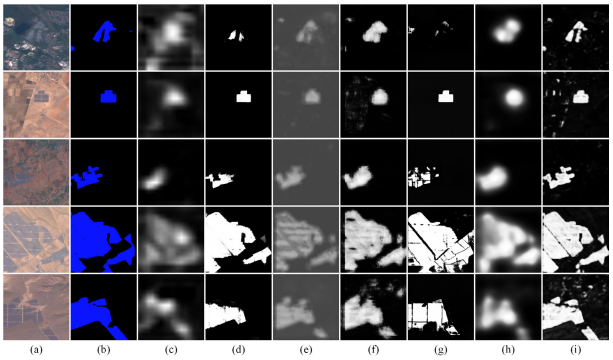


Fig. 14. Predicted mapping results of competing methods and our method on the GloSoFarID dataset. (a) Test images. (b) GT. (c) GradCAM. (d) MFNet. (e) UASR. (f) SP-RAN. (g) WeSAM. (h) PLNM. (i) CJLF (ours).

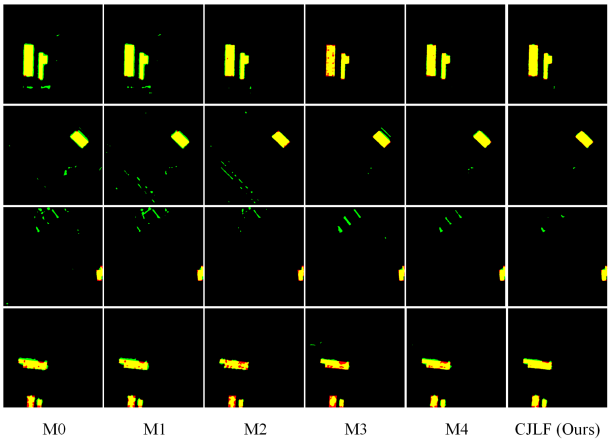


Fig. 15. Visual comparison of the proposed CJLF and its degraded version M0–M4 on the AerialGDA2020 dataset. True positives, true negatives, false positives, and false negatives are marked in yellow, black, green, and red, respectively.

to train the M1 model, respectively. Quantitative results in Table IV show that PLs generated by GradCAM achieve a good balance between precision and recall. Furthermore, through the refinement stage involving the M1 model, the segmentation quality is significantly improved, as evidenced by substantial improvements in both F_1 and mIoU metrics across both datasets. In contrast, while CAM w/Ref produces PLs that underperform overall compared to GradCAM, they excel in maximizing the identification of target objects,

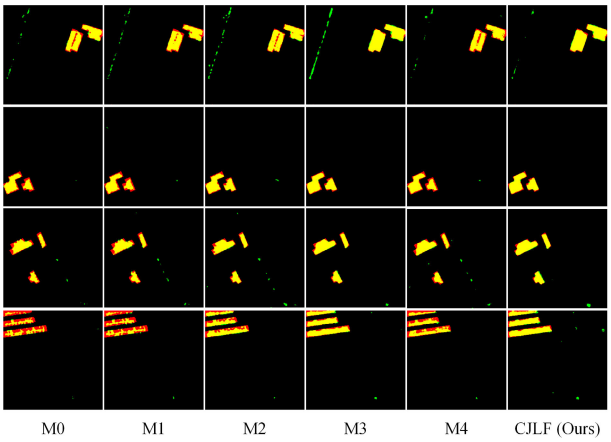


Fig. 16. Visual comparison of the proposed CJLF and its degraded version M0–M4 on the DualGMS dataset. True positives, true negatives, false positives, and false negatives are marked in yellow, black, green, and red, respectively.

reflected by their notably higher recall values. After the refinement stage, however, the overall performance sees a decline, which indicates that PLs produced by CAM w/Ref are likely involved with high-level label noise. Benefiting from the joint supervision of GradCAM and CAM w/Ref, we observe a significant improvement in overall performance. These observations support the fundamental assumption of our method: the two distinct PL types capture complementary information in terms of precision and recall. By leveraging their complementary strengths, CJLF successfully mitigates the limitations inherent to each individual PL source, thereby improving the overall robustness and accuracy of the model.

E. Parameter Analysis

For the loss function utilized to train CJLF [the objectives in (1) and (13)], we conduct experiments to study the effect of α, β, and γ on AerialGDA2020 and DualGMS datasets. α adjusts the contribution of the OLS to the training of CJLF. As shown in Fig. 17, α has a significant impact on the performance of CJLF. On the AerialGDA2020 dataset, the F_1 score increases significantly and reaches its peak when $\alpha = 0.3$, followed by a slight decline as α grows further. A similar trend is observed on the DualGMS dataset, though the performance reaches its maximum at $\alpha = 0.5$ before experiencing a subsequent decrease. β controls the impact

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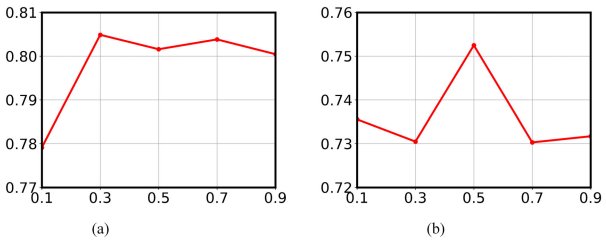


Fig. 17. F_1 score with α varying from 0.1 to 0.9 on (a) AerialGDA2020 and (b) DualGMS datasets.

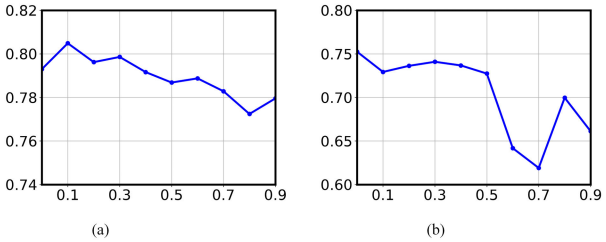


Fig. 18. F_1 score with γ varying from 0.1 to 0.9 on (a) AerialGDA2020 and (b) DualGMS datasets.

TABLE V
MIOU AND F_1-SCORES ON THE AERIALGDA2020 DATASET USING VARYING β

β	0	0.0001	0.001	0.01
F_1	0.7782	**0.8049**	0.7970	0.7929
$mIoU$	0.6609	**0.6882**	0.6782	0.6714

of the SCC, which helps reduce the ambiguity between the foreground objects and their visually similar co-occurrence. As shown in Tables V and VI, F_1 and mIoU reach peaks when $\beta = 0.0001$ on AerialGDA2020 and $\beta = 0.0005$ on DualGMS datasets. γ controls the involvement of pixels where PLs disagree in the training process of CJLF. Fig. 18 shows a consistent decrease in overall performance on both datasets as γ grows. On AerialGDA2020, the optimal empirical value for γ is 0.1, while the value is 0 for the DualGMS dataset. This also complies with our assumption that the pixels where the two types of PLs do not agree harm the model learning, and it is beneficial to leave these pixels to $\mathcal{L}_{\text{scc}}$. The parameter K was selected empirically through an experimental process. The purpose of selecting K pixels from the mask $\mathcal{M}$, and $\mathcal{M}_d$, rather than using the entire masks, is to mitigate the adverse effects of noisy pixels as well as reduce the computational cost of $\mathcal{L}_{\text{scc}}$. Considering the domain knowledge and computational burden, we defined a rough and reasonable range of K values to test, i.e., $K = [600, 800, 1000, 1200]$ for AerialGDA2020 and $K = [300, 500, 700, 900]$ for DualGMS. Table VII shows quantitative performance using each candidate K value. The results clearly demonstrate that performance peaks around our chosen value of $K = 1000$ for AerialGDA2020 and 500 for DualGMS.

TABLE VI
MIOU AND F_1-SCORES ON THE DUALGMS DATASET USING VARYING β

β	0	0.0005	0.001	0.0015	0.002
F_1	0.7454	**0.7525**	0.7413	0.7489	0.7477
$mIoU$	0.6292	**0.6378**	0.6250	0.6305	0.6336

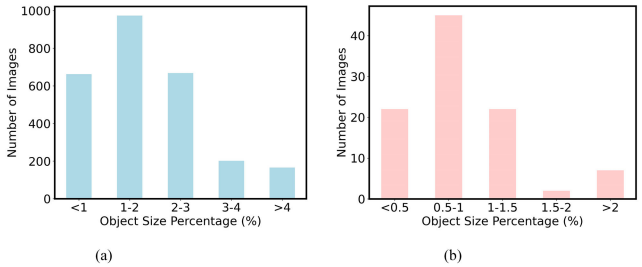


Fig. 19. Distributions of images with different object percentages in (a) AerialGDA2020 and (b) DualGMS test sets.

F. Performance on Objects With Varying Sizes

Considering the varying sizes of the targets, we study the distribution of the proportion of solar panels in the images on the AerialGDA2020 and DualGMS datasets. And conduct a subset performance evaluation. Specifically, concerning the proportion of solar panels in a single image, the test images are classified into several groups. For AerialGDA2020 dataset, we have five groups: 0%–1%, 1%–2%, 2%–3%, 3%–4%, >4%; for DualGMS dataset, we have 0%–0.5%, 0.5%–1%, 1%–1.5%, 1.5%–2%, >2%. From Fig. 19, it can be seen that the majority of solar panels account for less than 3% and 1.5% of an image in AerialGDA2020 and DualGMS test sets, respectively. Two adopted PLs generated by GradCAM and CAM w/Ref, WS-SOD, UASR, SP-RAN, and the proposed CJLF are selected to show the subset performance, which is summarized in Fig. 20.

Fig. 20 illustrates the challenge all methods face in detecting very small objects within complex backgrounds. Among these methods, GradCAM and CJLF demonstrate the strongest performance. GradCAM and CAM w/Ref exhibit lower IoU scores, particularly in higher object percentage ranges. This suggests a limitation in their ability to detect instances holistically. WS-SOD and SP-RAN consistently perform well, especially as the object percentage increases. Compared to UASR and SP-RAN, CJLF shows significantly reduced variance across different size groups and consistently delivers reliable detection capabilities, particularly for small-sized objects.

G. Limitations

While CJLF has shown encouraging results, some limitations need further investigation. One of the sensitive issues arises from the dramatic variation in object sizes commonly observed in remote sensing imagery. In particular, the majority of target objects, such as solar panels, often occupy only a small fraction of the overall image. This leads to a severe foreground–background imbalance, where small objects are

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TABLE VII
QUANTITATIVE RESULTS USING VARYING K ON AERIALGDA2020 AND DUALGMS DATASETS

Data set	K	AC	P	R	F_1	$mIoU$	MSE	E
AerialGDA2020	600	0.9916	**0.8419**	0.8004	0.7981	0.6865	**0.0134**	0.9076
	800	0.9910	0.7848	**0.8380**	0.7912	0.6733	0.0199	0.9163
	1000	**0.9929**	0.8076	0.8346	**0.8049**	**0.6882**	0.0137	**0.9252**
	1200	0.9911	0.8300	0.7781	0.7793	0.6607	0.0161	0.9028
DualGMS	300	0.9844	0.8359	0.7086	0.7460	0.6336	0.0250	0.8954
	500	**0.9851**	**0.8766**	0.6916	**0.7525**	**0.6378**	**0.0222**	0.8947
	700	0.9843	0.8610	0.6802	0.7359	0.6239	0.0236	0.8768
	900	0.9843	0.8359	**0.7092**	0.7486	0.6349	0.0255	**0.9017**

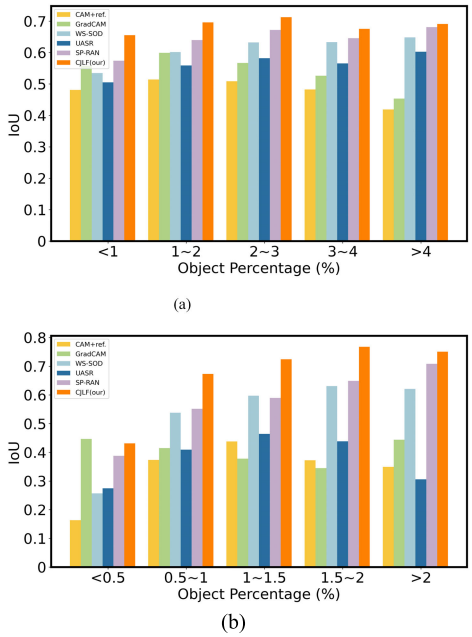


Fig. 20. Performance comparison of state-of-the-art methods across different object size ranges.

easily overlooked by vast and complex background regions. Although CJLF takes advantage of complementary PLs to enhance the identification of extremely small objects, further improvement in accurately segmenting and localizing such small-scale targets remains an important direction for future work.

Another limitation of the current CJLF framework is its reliance on only two types of PLs for supervision. While these complementary PLs help alleviate individual inductive biases, they represent a limited view. In real-world scenarios, incorporating a broader set of PLs derived from different annotation modalities (e.g., bounding boxes or point-level annotations) could provide more informative guidance for training. Extending CJLF to accommodate more heterogeneous PLs types is a valuable direction for future research.

V. CONCLUSION

Automatic SPM enables low-cost and scalable monitoring of the adoption of solar energy by leveraging high-resolution remote sensing imagery and machine learning techniques. In this article, we propose a CJLF for weakly supervised SPM, designed to fully leverage the complementary strengths of PLs while mitigating inductive biases. The architecture of the CJLF comprises an RA-based encoder and a dual-stream decoder. The RA-based encoder focuses on generating discriminative feature embeddings, whereas the dual-stream decoder operates with two specialized branches: the OSS and the OLS. The complementary learning is realized in two critical ways. First, the AFM fosters mutual reinforcement between the two branches to enhance each other's performance. Second, recognizing the challenge posed by ambiguous regions during joint training of segmentation and localization tasks, we introduce a subspace contrast loss, which helps distinguish target objects and ambiguous regions. To reveal the validity of the proposed CJLF, comprehensive experiments are conducted on three challenging datasets for solar panel mapping, including two aerial-image datasets, AerialGDA2020 and DualGMS, and one satellite-image dataset, GloSoFarID. Both visual and quantitative comparisons of the proposed method against the state-of-the-art WSL methods consistently demonstrated the effectiveness and superiority of CJLF.

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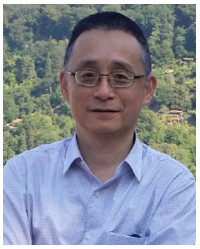
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